Lab05. Running container in a cloud

Creation and Administration of Cloud-based Applications

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Fall 2026

1 Learning objectives

1. Learn what is *Elastic Container Service* (ECS).
2. Use ECS service to run container from container image stored in ECR.

2 Exercises

2.1 Start

1. Start your AWS Learning Lab, and open AWS Console.
2. Start development environment created in the previous lab.

2.2 Create ECS cluster

1. Open the *Amazon Elastic Container Service* in AWS Console.
2. In the navigation pane (on the left), choose Clusters.
3. On the Clusters page, choose Create cluster.

   Warning: Ignore messages about problems with retrieving a list of namespaces.
4. Under Cluster configuration, for Cluster name, enter `<INITIALS>-cluster-01`. Replace `<INITIALS>` with your initials, eg. `js-cluster-01` for John Smith. The name can contain up to 255 letters (uppercase and lowercase), numbers, and hyphens.
5. Turn on Container Insights: expand Monitoring, and then turn on Use Container Insights.
6. Choose Create.

   Warning: If the creation fails, repeat procedure from the start.

2.3 Create a Task definition

1. In the navigation pane (on the left), choose Task definitions.
2. Click Create new task definition, and select Create new task definition.
3. Type Task definition family: `<INITIALS>-flask-app`
4. Change:
   - memory to 2GB
   - Task role to LabRole
   - Task execution role to LabRole

- in Container-1 configure info type:
  - Name - `flask-app`
  - Image URI - copy URI from ECR, it looks *similar* to: `051548163354.dkr.ecr.us-east-1.amazonaws.com/flask-app:latest`. Remember to copy URI of image, and **not** URI of ECR repository!
- Port mappings - set Container port to 5000.

5. Expand Monitoring section and select both checkboxes.
6. Click Create.

2.4 Create an ECS cluster service

1. In the navigation pane (on the left), choose Clusters, then select your cluster.
2. Select Services tab. Click Create.
3. From the list Family select your task definition.
4. In Service name field type: `<INITIALS>-flask-app-svc`.
5. Expand Load balancing:
   - Set Load balancer type to Application Load Balancer.
   - Select Create a new load balancer.
   - Set Load balancer name to: `<INITIALS>-flask-app-lb`.
   - Verify if Choose container to load balance shows your task.
   - In Target group name type: `<INITIALS>-flask-app-tg`.
6. Click Create.

2.5 Create a security group to access load balancer

1. Open the EC2 Service in AWS Console.
2. In the navigation pane (on the left), choose Security Groups.
3. Click Create security group.
4. Configure a new security group:
   1. Security group name: `webaccess`.
   2. Description: `HTTP access from anywhere`.
   3. In Inbound rule click Add new:
      1. Type: `HTTP`.
      2. Source: `Anywhere-IPv4`.
   4. Click Create security group.
5. In the navigation pane, choose Load Balancers.
6. Select Load Balancer `<INITIALS>-flask-app-lb`.
7. In the tab Security click Edit.
8. From Security groups drop-down list select group `webaccess`. Do not uncheck `default`!
9. Click Save changes.

2.6 Display your application

1. Open the EC2 Service in AWS Console.
2. In the navigation pane, choose Load Balancers.
3. Select your load balancer.
4. Copy DNS name.
5. Paste the DNS name in the browser and open the web site.

   You should be able to see your Flask application running.